INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2022-23

Date of Examination: 20 April 2023 **Session: AN** **Duration: 3 hrs.** **Full Marks: ___________100**

Subject No.: ___AI61004___Subject: __________STATISTICAL FOUNDATIONS OF AI/ML___________

Department/Center/School:__CENTRE OF EXCELLENCE IN ARTIFICIAL INTELLIGENCE________

Specific charts, graph paper, log book etc., required ______________________NO____________________

Special Instructions (if any): WRITE SHORT BUT PRECISE ANSWERS SHOWING ALL MATHEMATICAL DETAILS.

ANSWER ANY 5 OF THE QUESTIONS BELOW. ALL PARTS OF EACH QUESTION MUST BE ANSWERED TOGETHER

Q1: A) There is a Gaussian distribution N(u, 9). Consider a prior distribution N(0,25) on 'u'. You receive data sequentially from the Gaussian distribution [5.1, 4.3 7.7 5.4 2.1 4.8]. Show how the posterior and MAP estimation of 'u' evolve as this data arrives one by one? Derive the necessary expressions.

[10 marks]

B) Prove that if Y follows chi-squared(n), then E(Y)=n and Var(Y)=2n. **[5 marks]**

C) Given an example of an estimator which is biased but asymptotically unbiased, with necessary derivation.

[5 marks]

Q2: A) Prove that the samples obtained by rejection sampling actually follow the target distribution. **[5 marks]**

B) You want to calculate the expectation of a Gamma Distribution Ga(4,3) using Importance Sampling using a suitable Gaussian distribution. Using the following pseudorandom number sequence in (1,100), demonstrate the process using 10 samples.

Random number sequence: [82 91 13 32 64 10 28 55 96 47] **[8 marks]**

C) What are the pros and cons of the proposal distribution having high variance in case of Metropolis-Hastings?

[4 marks]

D) Explain the significance of "burn-in phase" in MCMC methods. **[3 marks]**

Q3. A) You have drawn 3 samples x1 ~ Gamma(a,1), x2 ~ Gamma(b,1), x3 ~ Gamma(c,1). Consider y=(y1,y2,y3) where yi=xi/(x1+x2+x3). Prove that y ~ Dirichlet(a, b, c). **[8 marks]**

B) Discuss how we can carry out Gibbs Sampling to estimate the latent variables of a GMM. You can consider all the Gaussian component means (μ1, μ2, … μK) as latent variables following a

Gaussian prior $N(0, s^2)$ and the proportion Π of mixture components as following a Dirichlet prior with parameters ($\alpha 1$, $\alpha 2$, ... αK). The Gaussian mixture variances ($\sigma 1$, $\sigma 2$ ... σK) can be considered as equal and known (σ). Show the joint distribution and the conditionals required for Gibbs Sampling. **[12 marks]**

Q4. A) Consider the following model: $Y_i \sim Ber(p)$, $Z_i \sim U(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$

We have 3 observations of X and Z: [X1=3.6 Z1=0.2, X2=-1.2 Z2=0.5, X3=4.2 Z3=0.8]

Consider the following 2 choices of parameters:

P1: [W1=-5, W2=5, s=9, p=0.5], P2: [W1=0, W2=4, s=25, p=0.8]

Which of these two sets of parameters fits the observations better? We do not observe Y. **[8 marks]**

B) Show that E-M algorithm for GMMs is analogous to K-means clustering under certain parameter settings. Can you suggest any change to K-means to make it more similar to E-M? (hint: compare the E-M steps and K-means steps) **[8 marks]**

C) In which circumstances can E-M lead to wrong estimates of parameters of a GMM? **[4 marks]**

Q5. A) How can you compare an Auto-Regressive model and an HMM on the given sequential data? **[6 marks]**

Data sequence X=[-2.2, 0.9, 1.3, -1.8]

AR model: X(1)~N(0,4), X(t+1)~N(1.1*X(t),4)

HMM model (hidden states S1,S2): $\pi = [0.5\ 0.5]$ (Initial state distribution)

A(S1,S1)=A(S2,S2)=0.8 (Transition Distribution)

X(t)|Z(t)=S1 ~ N(-2,1), X(t)|Z(t)=S2 ~ N(1,1) (Emission Distribution)

B) You have a dataset X of 10 binary observations. Your Null Hypothesis is that the Bernoulli parameter is 0.5, while the alternate hypothesis is that this parameter is below 0.5. Your test-statistic T(X) is the fraction of '1's, and your rejection region is defined by ($T(X)-0.5<\epsilon$). What will be possible values of ϵ, if you are to reject the null hypothesis at 5% level of significance? **[4 marks]**

C) You have two datasets: A=[2.3 3.1 2.8] and B=[3.0 3.6 5.2]. The null hypothesis is that, they follow the same distribution. Test this hypothesis using the Permutation test over any 10 (randomly chosen) permutations, at 5% level of significance. **[5 marks]**

D) A dataset of 20 observations are divided into 5 groups of 4 observations each. The maximum values of these groups are x1, x2, x3, x4, x5. If the data are known to follow a distribution for which the 90th percentile is 'c', what is the expected number of groups for which the group maximum value will exceed 'c'? **[5 marks]**

Q6. A) Draw 10 samples X= (x1, x2, ... x10) from Beta(5,10), using rejection sampling (you can ignore the denominators of Beta PDF for sampling). Use the pseudorandom number sequence below for this purpose. **[4 marks]**

B) Using each of the above sampled values, draw samples Y=(y1,y2,....y10) such that yi~N(xi,4) using the same pseudorandom number sequence. **[4 marks]**

C) Fit Kernel-based Density-based functions f(X) and g(Y) on these samples using Gaussian Kernels. For both X and Y, write the expressions of the fitted density functions at any arbitrary point. **[2 marks]**

D) Calculate the difference between the original and fitted distributions (both X and Y) at 3 uniformly spaced points within the range of the samples. **[4 marks]**

E) Discuss, without numerical calculations, how you can use a Copula Function C(θ) to estimate a joint density h(X,Y) using the fitted functions f(X) and g(Y). **[4 marks]**

F) Finally, how will you evaluate the quality of the fitted joint distribution h(X,Y) using a suitable test of hypothesis? **[2 marks]**

Random number sequence: [82 91 13 92 64 10 28 55 96 97 16 98 72 49 81 15 43 22 80 37 66 4 85 94 68 76 75 40 66 18]